The NSGA planner: what it computes, line by line

A typeset walkthrough of the mathematics, the code and the numbers

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1 Takeaways and what changed after review

If you read only this box.

1. **What it is.** Scores 4,284 strategies $x = (\omega, s, r)$ (rotation, fertilizer scale, stover retention) on five objectives by interpolating 2,016 WOFOST crop-simulation seasons; NSGA-II/III (evolutionary multi-objective search) is checked against brute force over all 4,284.
2. **The arithmetic is right.** An independent pandas recomputation of 10 strategies matches the planner to a relative 1.2×10^{-16}; all 35 unit tests pass.
3. **Two inputs set the conclusions, not the algorithms:** how much yield nitrogen buys (calibrated WOFOST: none, $2.301 \to 2.312\,\mathrm{t\,ha^{-1}}$; default: almost linear) and placeholder prices (the faba bean price used is 3.3× SIAP's latest value).
4. **"N lost" restates the dose:** $\approx 0.49\,D$ in the default world (by the fixed 50% recovery constant), $\approx 0.95\,D$ in the calibrated one. It is not a measured loss.
5. **The big fronts are partly artifacts:** 483 calibrated strategies come from dominance resistance (Section 5). A float-noise bug had kept 21 more on the front (504 before) and produced one of the picks; it is fixed (Section 5.4).
6. **NSGA-III never "plateaued" at 89–94%:** that was a reporting bug (`result.X`); the final population covers 95–98.5% of the hypervolume, and NSGA saves no evaluations over brute force here.
7. **Chebyshev picks and the robust "hedge dose" are unstable.** The hedge (48–$132\,\mathrm{kg\,N\,ha^{-1}}$) holds only under range-normalized regret (176–200 in absolute pesos).
8. **Resilience is weak evidence.** Bad-year income is the mean of the worst 4 of 16 years, which are frost-truncated seasons (correlation with yield -0.70), not droughts; the holdout cannot test drought; the SMAP agreement ($r = 0.87$) feeds nothing and could not be re-run.

Scope. This document explains `src/crop_models/nsga_planner.py` (multi-objective planning of rainfed rotations at Texcoco, Mexico) as it stands in the working tree on October 26, 2026 (HEAD `21e30ec` plus uncommitted changes), replaces the plain-text draft `src/crop_models/NSGA_WALKTHROUGH.md` (keeping its structure and status table) and is meant to be read alone. Every number was recomputed from `data/nsga/grid.csv` or the saved fronts in `data/nsga/*/` with the repository's own code (Section 10). Statements are labeled *code* (what the code does), *docs* (what repository documents say) or *my reading* (interpretation).

Status of the earlier review findings

The first review read the code just before these changes. "Fixed" means I confirmed the fix in the current code or data.

Finding	Status
NSGA-III's "89–94% plateau" is a reporting artifact: pymoo's `result.X` keeps one strategy per occupied reference direction.	**Fixed.** `optimize` returns the non-dominated set of the final population `result.pop` (`nsga_planner.py:323--326`; test `tests/test_nsga_planner.py:199`); coverage 95–98.5% (Table 16). The "thin front" explanation is removed.
"Worst 20% of years" is the worst 25% for 16 years ($q = \lceil 0.2 \cdot 16 \rceil = 4$).	**Fixed by relabeling** to the worst quarter, `CVAR_SHARE = 0.25` (`nsga_planner.py:82`). No number changes.
Default-world stover and roots are not capped like grain.	**Documented**, not changed (shared simulator code); still open (Section 4).
Doc slips ("3 kg" should be 4; Reproduce commands did not match `run.json`).	**Fixed.** The saved runs reproduce here (Section 6).
Latest SIAP faba bean price (2025: 5,030 MXN t^{-1}) is 0.30× the price used.	**Documented.** The price used is the median of 16,548, 16,524, 5,030.
The minimax dose depends on how regret is normalized (176–200 kg N ha^{-1} in absolute pesos).	**Verified and documented**; derived in closed form in Section 8.
"Benign years" for 2022–2025 is unsupported; "N loss is a lower bound" unverifiable; IGBP's carbon stock is read by no objective.	**Rewritten** in the results document.
Calibrated picks sit inside the first interpolation cell $[0, 0.25]$, whose interior was never simulated.	**Documented**; a finer low-dose grid is a next step. After the float fix the calibrated picks are at $s = 0$ (a simulated node), but the 231 maize front members still lie inside the cell.
21 calibrated front members existed only through a 4.4×10^{-16} t ha^{-1} float difference (second review); the first review's "no pick changes" was wrong.	**Fixed.** `pareto_front` (`nsga_planner.py:231--244`) compares objectives rounded to 9 decimals (`FRONT_DECIMALS = 9`; tests `tests/test_nsga_planner.py:163,172`). Front sizes, the coverage tables and the calibrated sustainability-first pick changed (Section 5.4).

Table 1. Findings of the first review and their status.

Corrections made while re-typesetting this document

Every number of the Markdown draft was re-verified; six changed (marked where they occur).

Item	Markdown draft	Re-verified value
Maize–faba bean 2018 N-loss example	"the faba phase took up 97.5 kg"	It is the *maize* phase ($Q = 97.52$, $Q(0) = 88.54$, $D = 80$); the faba phase has $D = 20$ and no extra uptake.
Front members dominated once yield is dropped	374 of 462 (leaving 88)	444 of 462 (leaving 18)
Standing mineral N versus "lost" N (calibrated, maize, $s = 1$)	"about 3% of the 152 kg lost"	68.7 kg N ha^{-1} more mineral N at the start of the 2021 season than at $s = 0$, i.e. 2.8% of the 16-year cumulative "loss" $16 \times 152.2 = 2{,}435$ kg N ha^{-1}
Crossing of the two income-shortfall lines	$N^* = 0.180/(a + b) = 87.5$	87.3 by the closed form; 89.4 exact on the piecewise-linear curves; lattice optimum 88
Held-out net income of the two picks	11,598 vs 6,086 (picks of an earlier front)	11,598 vs 6,158 for the picks on the current saved front
Faba bean area behind the price	"4–12 ha"	10–12 ha of faba bean (the 4–12 range is barley and faba bean together)

Table 2. Numbers corrected relative to `NSGA_WALKTHROUGH.md`.

2 The picture and the pipeline

The question. A farmer at Texcoco (Valley of Mexico, 19.53°N, 98.85°W, rainfed, one crop per rainy season) can choose a rotation, a fertilizer level and how much stover (the leaves and stems left after harvest) to leave in the field. Which choices are good *jointly* for yield, income, nitrogen lost to the environment, soil carbon returned and surviving bad years? There is no single best answer, so the planner returns the *Pareto front*: the strategies that no other strategy beats on every objective at once.

What goes in and out. In: a table of 2,016 simulated crop seasons (`data/nsga/grid.csv`) from WOFOST (WOrld FOod STudies, a daily crop growth model; run through PCSE, the Python Crop Simulation Environment) for two parameter sets, plus prices and costs. Out: for every strategy five numbers, the Pareto front of those numbers, and (optionally) one compromise pick. Where it sits: it is the last stage of the pipeline, downstream of ingest, simulation and calibration.

NASA data → WOFOST runs → `grid.csv` → `Grid` (arrays) ↓

`choose` / `RobustPlanner` ← NSGA-II/III + brute force ← `Planner.objectives` ← interpolation along s

The planner chain in code: `Grid` (`nsga_planner.py:106`) → `Planner.annual` (:164) → `Planner.objectives` (:182) → `brute_force` (:258), `optimize` (:266), `front_quality` (:329), `choose` (:348), `RobustPlanner` (:190).

2.1 Which dataset feeds which stage

Dataset	What it provides here	Where used
POWER (NASA Prediction Of Worldwide Energy Resources)	Daily weather 2005–2025 at the site: radiation, temperature, rain. Drives every simulated season; its year-to-year spread is the only source of risk in the bad-year objective.	`wofost_rotations.py:342`
IGBP-DIS (soil data of the International Geosphere–Biosphere Programme, via NASA Earthdata)	Field capacity 0.373 and wilting point 0.137 $\mathrm{m^3\,m^{-3}}$ (0–100 cm); total soil N 12,709 $\mathrm{kg\,N\,ha^{-1}}$, of which 0.5% per season is mineralized: $0.005 \times 12{,}708.6 = 63.5\,\mathrm{kg\,N\,ha^{-1}}$	`wofost_rotations.py:149-150,157,118`
GLDAS (Global Land Data Assimilation System)	Soil porosity 0.437 $\mathrm{m^3\,m^{-3}}$	`wofost_rotations.py:151`
IGBP-DIS soil carbon	130.4 $\mathrm{t\,C\,ha^{-1}}$ stock, written to `grid.json` only. *No objective reads it.*	`nsga_planner.py:526-530`
SMAP (Soil Moisture Active Passive)	An earlier validation of simulated soil water (below); *not used by the planner.*	`wofost_rotations.py:391`
SIAP (Mexican agriculture ministry; not NASA)	Prices (median of the last three priced years: maize 7,981, barley 5,240, faba bean 16,524 $\mathrm{MXN\,t^{-1}}$) and the observed yields behind the calibration.	`farm_tool.py:98`
Placeholders (no source)	Other costs (14,000, 9,000, 11,000 $\mathrm{MXN\,ha^{-1}}$), N price 25 $\mathrm{MXN\,(kg\,N)^{-1}}$, stover price 1,000 $\mathrm{MXN\,t^{-1}}$, carbon fraction 0.45.	`nsga_planner.py:78-81`

Table 3. Inputs of the planner and their provenance. The grid has 8 rotation-phases ($1+2+2+3$) × 21 years × 6 fertilizer scales × 2 parameter sets = 2,016 rows; the planner uses 2006–2021 (design) and 2022–2025 (holdout).

2.2 The SMAP check: what it is and what it is not

Code (`wofost_rotations.py:391--402`): the simulated 0–100 cm soil water WWLOW (cm) is divided by the rooting depth of 100 cm to give a volumetric fraction $\mathrm{SM}_d^{\mathrm{sim}}$ ($\mathrm{m^3\,m^{-3}}$), and compared with SMAP's L4 root-zone moisture $\mathrm{SM}_d^{\mathrm{SMAP}}$ (daily means of 3-hourly values) by a Pearson

correlation over the overlapping days d,

$$r = \mathrm{corr}_d\big(\mathrm{SM}_d^{\mathrm{sim}},\ \mathrm{SM}_d^{\mathrm{SMAP}}\big), \qquad \mathrm{SM}_d^{\mathrm{sim}} = \frac{\mathrm{WWLOW}_d}{100\ \mathrm{cm}}. \tag{1}$$

Docs: $r = 0.87$ over 731 days (2023–2024) for the maize-every-year rotation (`docs/technical_report/tables/smap.tex`). **I could not re-run this**: it needs a PCSE simulation, which was excluded here; I also cannot tell from the repository which parameter set (default or calibrated) produced it.

What it is not (*my reading*): it calibrates nothing, and `nsga_planner.py` never reads SMAP. Pearson r ignores bias and scale. The 731 daily values are not independent (the lag-1 autocorrelation of the SMAP series in `data/smap_texcoco_2023_2024.csv` is 0.998, lag 30 is 0.83; I computed this), so r is dominated by the shared seasonal wet–dry cycle and the effective sample is far smaller than 731. And it compares one simulated field with a ~ 9 km SMAP cell. It says the simulated water follows the right seasons, not that drought years are reproduced.

2.3 Notation used throughout

Symbol	Meaning	Unit or value
ω	rotation (categorical): maize, maize–barley, maize–faba bean, maize–barley–faba bean	—
s	fertilizer scale, a multiple of each crop's dose ($s = 1$: maize 160, barley 60, faba bean 40)	dimensionless, $[0, 1.25]$
r	stover retention, the share of stover left in the field	fraction, $[0, 1]$
$x = (\omega, s, r)$	a strategy; the lattice is Ω, $\lvert\Omega\rvert = 4{,}284$	—
$N = 160\,s$	"maize N", the maize dose implied by s	kg N ha^{-1}
$t,\ T;\ u,\ n_u$	design year (2006–2021, $T = 16$); phase of the rotation ($n_u = 1, 2, 2, 3$)	—
c_{tu}	crop grown in year t in phase u	maize, barley, faba bean
G, D, Q, S, B	grain, N dose, N uptake, stover (leaves incl. dead + stems), roots, per season	t ha^{-1} dry matter, kg N ha^{-1}, kg N ha^{-1}, t ha^{-1}, t ha^{-1}
π_c, K_c; π_N, π_S, ϕ	grain price, other cost of crop c; N price 25, stover price 1,000, carbon fraction 0.45	MXN t^{-1}, MXN ha^{-1}; MXN (kg N)$^{-1}$, MXN t^{-1}, t C (t DM)$^{-1}$
Y_t, V_t, L_t, C_t	annual (phase-averaged) yield, net income, N loss, carbon input	t ha^{-1}, MXN ha^{-1}, kg N ha^{-1}, t C ha^{-1}
$f_1, \ldots, f_M$; σ_i	objectives in minimization form $f_i = \sigma_i \times$ natural value, $\sigma_i = \pm 1$; $M \in \{2, \ldots, 5\}$	—
$\alpha,\ q$	tail share 0.25 and tail count $q = \lceil \alpha T \rceil = 4$	—
θ	WOFOST parameter set ("world"): default (def) or calibrated (cal)	—

Table 4. Symbols. A symbol keeps its meaning for the whole document.

3 The grid and linear interpolation along the fertilizer scale

Plain language. WOFOST was run only at six fertilizer levels. A strategy at a finer level, say $s = 0.6$, is scored by blending the two neighboring simulated levels, season by season, as if yield were a straight line between them. This is what makes a strategy cost about 60 μs instead of a 3–10 s simulation.

3.1 The equation

The grid nodes are $s_0, \ldots, s_5 = 0,\ 0.25,\ 0.5,\ 0.75,\ 1,\ 1.25$ (`GRID_SCALES`, `nsga_planner.py:60`). For any simulated quantity $X \in \{G, D, Q, S, B\}$ of one (year, phase) cell, a strategy with scale s (clipped to $[0, 1.25]$) uses

$$j(s) = \min\big\{4,\ \max\{j :\ s_j \le s\}\big\}, \qquad w = \frac{s - s_j}{s_{j+1} - s_j}, \qquad X(s) = (1 - w)\,X_j + w\,X_{j+1}. \tag{2}$$

Here $w \in [0,1]$ is the weight on the upper node (dimensionless) and X_j is the value simulated at node s_j. Code: `Grid.interpolate`, `nsga_planner.py:136--142`; the clip is line 138, j is line 139, w is line 140 and the blend is line 141. At a node $w = 0$ and the table value is returned exactly (test `tests/test_nsga_planner.py:63`).

Worked number (default parameters, pure maize, 2006, $s = 0.6$). Here $j = 2$ ($s_2 = 0.5$) and

$$w = \frac{0.6 - 0.5}{0.75 - 0.5} = 0.4, \qquad G(0.6) = 0.6 \times 5.965005 + 0.4 \times 7.993159 = 3.579003 + 3.197264 = 6.776267 \text{ t ha}^{-1}. \tag{3}$$

The code returns 6.776266824; hand and code agree.

3.2 What the nodes look like

		fertilizer scale s_j (maize dose $D = 160\, s_j$)					
Parameters	Mean over 2006–2021, pure maize	0	0.25	0.5	0.75	1	1.25
default	yield $\bar{G}$ [t ha^{-1}]	1.918	3.038	4.162	5.252	6.482	7.599
default	uptake $\bar{Q}$ [kg N ha^{-1}]	66.6	86.8	107.0	127.2	147.4	167.6
calibrated	yield $\bar{G}$ [t ha^{-1}]	2.301	2.312	2.312	2.312	2.312	2.312
calibrated	uptake $\bar{Q}$ [kg N ha^{-1}]	59.9	67.3	67.8	67.8	67.8	67.8

Table 5. Node means of grain yield $\bar{G}$ and N uptake $\bar{Q}$ (pure-maize rotation, design years).

Default world: a near-linear nitrogen response. Uptake rises 20.2 kg N ha^{-1} per 40 kg N ha^{-1} of dose, an *apparent recovery*

$$\mathrm{AR}(s) = \frac{\bar{Q}(s) - \bar{Q}(0)}{\bar{D}(s)} \quad [\text{dimensionless}], \qquad \mathrm{AR}_{\mathrm{def}}(0.25) = \frac{86.816 - 66.584}{40} = 0.506, \tag{4}$$

which is the hand-set `N_RECOVERY = 0.5` (`wofost_rotations.py:117`, "share of fertilizer N the crop can access") plus a little carry-over. Yield rises by about 1.1 t ha^{-1} per node.

Calibrated world: no response, and why. Yield moves from 2.3008 to 2.3125 t ha^{-1} (+0.0117, +0.51%) and uptake saturates at 67.75 kg N ha^{-1} from $s = 0.5$ on, giving $\mathrm{AR}_{\mathrm{cal}}(1) = (67.75 - 59.93)/160 = 0.049$. *Docs* (`CALIBRATION_RESULTS.md`): the calibration absorbed the Texcoco yield gap into photosynthesis (maize scaled by 0.47, grid.json), so growth is limited by carbon rather than nitrogen. The crop's N demand tops out near 68 kg N ha^{-1} while the soil already supplies about 63.5 (mineralization above) plus carry-over, so extra fertilizer has nothing to do. *My reading*: this is a property of the fitted parameter set, whose only nitrogen evidence is one station trial (unfertilized/fertilized ratio 0.93; the model's is 0.995), not an agronomic finding. *Docs* on the fit: the calibrated knobs per crop are a thermal-time (phenology) scale and a photosynthesis scale (maize 0.8 and 0.47, `grid.json`; the 0.8 sits on the agronomic lower bound), with N recovery and soil mineralization left at their defaults; they were fit to the 16 yearly SIAP yields of 2006–2021, and 2022–2025 was held out.

3.3 Assumptions and where they break

- **Histories are mixed.** Each node is a 21-year simulation at a fixed s, chaining water and soil N from season to season. A season at $s = 0.6$ would carry a 0.6-history; Equation (2) blends the 0.5- and 0.75-histories instead.
- **Straight line between nodes.** Reasonable where the response is near-linear (default), not where it jumps (calibrated uptake between $s = 0$ and 0.25, a maximum change of 29 kg N ha^{-1} in one season).
- **The lattice is 10× finer than the nodes.** Strategies sit every 0.025, nodes every 0.25. The calibrated picks are at $s = 0$ (a simulated node), but the 231 maize strategies on the five-objective front (s = 0–0.25) rest on interpolation inside the first cell $[0, 0.25]$, whose interior was never simulated.

Leave-one-node-out error. Predict each interior node $j = 1, \ldots, 4$ from its two neighbors, $\hat{X}_j = \frac{1}{2}(X_{j-1} + X_{j+1})$, and compare with the simulated X_j over all rotations, years and phases. The spacing is twice the real one, so this overstates the error of Equation (2).

World	Quantity	mean $\lvert\hat{X}_j - X_j\rvert$ per season	95th pct	max
default	grain yield ($\mathrm{t\,ha^{-1}}$)	0.048	0.20	0.82
default	N uptake ($\mathrm{kg\,N\,ha^{-1}}$)	0.014	0.043	0.42
calibrated	grain yield ($\mathrm{t\,ha^{-1}}$)	0.000	0.000	0.064
calibrated	N uptake ($\mathrm{kg\,N\,ha^{-1}}$)	0.24	0.000	13.2

Table 6. Leave-one-node-out error. The error of the 16-year, all-phase *mean* is at most $0.069\,\mathrm{t\,ha^{-1}}$ for default yield (about 1%) and $3.42\,\mathrm{kg\,N\,ha^{-1}}$ for calibrated uptake.

4 The five objectives, and one strategy worked end to end

Plain language. For a given strategy the planner builds, for each of the 16 design years, four farm-level numbers (yield, net income, N lost, carbon input), averaging over the rotation's phases (a hectare is, on average, half maize and half faba bean each year in a two-crop rotation). The objectives are the 16-year means of those, plus a risk measure: the mean net income of the worst quarter of years.

4.1 The equations

For one (year, phase) cell, let G, D, Q, S, B be the interpolated values from Equation (2) at the strategy's s, and $Q(0)$ the N uptake of the *same cell* at the zero-fertilizer node. With crop $c = c_{tu}$ in that cell, the four annual numbers are phase averages (n_u phases):

$$Y_t = \frac{1}{n_u}\sum_{u=1}^{n_u} G_{tu} \qquad [\mathrm{t\,ha^{-1}\ dry\ matter\ grain}] \tag{5}$$

$$V_t = \frac{1}{n_u}\sum_{u=1}^{n_u}\Big[\pi_c\, G_{tu} - \big(K_c + \pi_N D_{tu}\big) + \pi_S\,(1-r)\,S_{tu}\Big] \qquad [\mathrm{MXN\,ha^{-1}}] \tag{6}$$

$$L_t = \frac{1}{n_u}\sum_{u=1}^{n_u}\max\Big\{0,\ D_{tu} - \big(Q_{tu} - Q_{tu}(0)\big)\Big\} \qquad [\mathrm{kg\,N\,ha^{-1}}] \tag{7}$$

$$C_t = \frac{1}{n_u}\sum_{u=1}^{n_u}\phi\,\big(B_{tu} + r\,S_{tu}\big) \qquad [\mathrm{t\,C\,ha^{-1}}] \tag{8}$$

The five objectives (all in natural units; the search minimizes $f_i = \sigma_i \times$ objective with $\sigma = -1$ for the maximized ones, `NATURAL_SIGN`, :69):

$$\bar{Y} = \frac{1}{T}\sum_{t=1}^{T} Y_t, \qquad \bar{V} = \frac{1}{T}\sum_{t=1}^{T} V_t, \qquad \bar{L} = \frac{1}{T}\sum_{t=1}^{T} L_t, \qquad \bar{C} = \frac{1}{T}\sum_{t=1}^{T} C_t, \tag{9}$$

$$\mathrm{CVaR}_\alpha = \frac{1}{q}\sum_{l=1}^{q} V_{(l)}, \qquad q = \max\{1, \lceil \alpha T \rceil\} = \lceil 0.25 \times 16 \rceil = 4, \qquad V_{(1)} \le V_{(2)} \le \cdots \le V_{(T)}. \tag{10}$$

CVaR_α (conditional value at risk, here of a gain, i.e. the expected shortfall of income) is the mean of the q lowest annual incomes $V_{(l)}$ (order statistics, $\mathrm{MXN\,ha^{-1}}$). Because $\alpha T = 4$ is an integer, this is exactly the textbook empirical estimator of the lower-tail mean at level α; for non-integer αT the ceiling makes the tail slightly larger than α.

Term	Code	Line
G, D, Q, S, B at scale s	`grain, dose, uptake, stover, roots = (g.interpolate(rotation, name, n_scale) for name in FIELDS)`	:174
$\max\{0,\ D-(Q-Q(0))\}$	`np.maximum(0.0, dose - (uptake - g.rotations[rotation]["n_uptake_kg_ha"][0]))`; node index 0 is $s=0$	:175
V	`grain*price - (cost + e.n_price*dose) + e.stover_price*(1-retention)*stover`	:176–177
C	`e.carbon_fraction*(roots + retention*stover)`	:178
phase average $\frac{1}{n_u}\sum_u$	`.mean(axis=1)` (axis 1 = phase), after the floor and the prices are applied cell by cell	:179–180
year mean $\frac{1}{T}\sum_t$	`.mean()` in `objectives`	:185–186
CVaR_α	`tail_mean(values, share=CVAR_SHARE)`: `k = max(1, int(np.ceil(share*len(values))))`; mean of `np.sort(values)[:k]`	:145–148, 187
π_N, π_S, ϕ, K_c	`PLACEHOLDER_*`, `CARBON_FRACTION`, `CVAR_SHARE`	:78–82

Table 7. Symbols to code in `nsga_planner.py`.

Assumptions. The crop in each (year, phase) cell is the same at every s (checked at construction, :131–132). Retention r is pure accounting: it moves stover between income (π_S) and carbon (ϕ) and changes nothing in the simulation. Prices are constants (no price series), and a phase-averaged hectare is a farm of equal-sized fields.

4.2 One strategy, end to end

Strategy: *calibrated* parameters, maize–faba bean, $s = 0.5$ (doses $D = 80$ for maize and 20 for faba bean, so $50\,\mathrm{kg\,N\,ha^{-1}}$ phase average), $r = 0.5$, design years 2006–2021. At $s = 0.5$, $w = 0$ in Equation (2), so the node values are used directly. Rotations are offset in time, so in odd years phase 1 is maize and phase 2 faba bean, and vice versa.

Step 1: one year by hand, 2007 (phase 1 is maize). Prices: maize 7,981.46, faba bean $16{,}524\,\mathrm{MXN\,t^{-1}}$.

	maize ($u=1$)	faba bean ($u=2$)	phase average
G ($\mathrm{t\,ha^{-1}}$)	2.1785	1.7079	1.9432
D ($\mathrm{kg\,N\,ha^{-1}}$)	80	20	50
$Q = Q(0)$ ($\mathrm{kg\,N\,ha^{-1}}$)	61.936	21.095	
S, B ($\mathrm{t\,ha^{-1}}$)	1.3774, 0.2170	0.8111, 0.2299	
revenue $\pi_c G$ ($\mathrm{MXN\,ha^{-1}}$)	$2.1785 \times 7981.46 = 17{,}387.5$	$1.7079 \times 16524 = 28{,}221.7$	
other cost K_c	$-14{,}000$	$-11{,}000$	
N cost $\pi_N D$	$-25 \times 80 = -2{,}000$	$-25 \times 20 = -500$	
stover income $\pi_S(1-r)S$	$1000 \times 0.5 \times 1.3774 = 688.7$	$1000 \times 0.5 \times 0.8111 = 405.6$	
net V	2,076.2	17,127.3	9,601.7
N loss $\max\{0, 80-0\}$, $\max\{0, 20-0\}$	80	20	50.0
carbon $0.45\,(B + 0.5\,S)$	0.4075	0.2860	0.3467

Table 8. The year 2007 by hand. The code returns $V = 9{,}601.73$, $L = 50.0$, $C = 0.34675$, $Y = 1.94321$ for this year (identical).

Step 2: all 16 years. Every year has dose $50\,\mathrm{kg\,N\,ha^{-1}}$; $L_t = 50$ except 2018.

Year t	phase-1 crop	Y_t (t ha^{-1})	V_t (MXN ha^{-1})	L_t (kg N ha^{-1})	C_t (t C ha^{-1})
2006	fababean	2.134	10,590	50.0	0.356
2007	maize	1.943	9,602	50.0	0.347
2008	fababean	1.692	7,547	50.0	0.319
2009	maize	2.413	13,499	50.0	0.401
2010	fababean	0.887	−1,727	50.0	0.256
2011	maize	1.212	3,695	50.0	0.376
2012	fababean	1.553	6,937	50.0	0.323
2013	maize	2.191	12,279	50.0	0.366
2014	fababean	2.252	13,213	50.0	0.384
2015	maize	2.526	15,015	50.0	0.427
2016	fababean	2.388	13,961	50.0	0.428
2017	maize	2.081	10,869	50.0	0.384
2018	fababean	2.724	17,375	45.5	0.480
2019	maize	2.570	15,494	50.0	0.422
2020	fababean	1.676	8,006	50.0	0.364
2021	maize	2.429	14,570	50.0	0.391
mean		2.042	10,683	49.72	0.3765

Table 9. Annual values for the worked strategy (code output; the 2007 row matches Table 8).

Step 3: the five numbers.

$$
\begin{aligned}
\bar{Y} &= 2.042 \text{ t ha}^{-1}, \qquad \bar{C} = 0.3765 \text{ t C ha}^{-1}, \\
\bar{L} &= \frac{15 \times 50 + 45.512}{16} = \frac{795.512}{16} = 49.72 \text{ kg N ha}^{-1}, \\
\bar{V} &= \frac{170{,}925.2}{16} = 10{,}682.8 \text{ MXN ha}^{-1}, \\
\mathrm{CVaR}_{0.25} &= \frac{V_{2010} + V_{2011} + V_{2012} + V_{2008}}{4} \\
&= \frac{-1{,}726.85 + 3{,}695.48 + 6{,}937.05 + 7{,}546.86}{4} = \frac{16{,}452.54}{4} = 4{,}113.14 \text{ MXN ha}^{-1}.
\end{aligned}
\tag{11}
$$

The code's `Planner.objectives("maize-fababean", 0.5, 0.5)` returns exactly these: yield 2.041985, net 10,682.826, N loss 49.7195, carbon 0.376539, bad-year income 4,113.135. The worst four years are 2010 (−1,727), 2011, 2012 and 2008.

4.3 The difference method for N loss, and what it over- or understates

Code. Equation (7) says: loss = dose minus the *extra* uptake caused by the fertilizer, where the no-fertilizer uptake $Q(0)$ is the same year and phase from the $s = 0$ node. In 2018 the *maize* phase took up 97.52 against 88.54 unfertilized, an extra 8.98, so its loss was $80 - 8.98 = 71.02$ and the year's average is $(71.02 + 20)/2 = 45.51$. In every other year the uptake is identical with and without fertilizer, so the loss equals the dose. (**Correction**: the Markdown draft attributed this to the faba bean phase.)

My reading of what the figure measures:

- **Default world: a restatement of an assumption.** The simulator lets the crop reach only 50% of applied N (`N_RECOVERY = 0.5`); the crop then takes up essentially all of that half. So $L \approx (1 - \mathrm{AR})\, D \approx 0.495\, D$: maize at $s = 1$ loses 79.17 of 160 kg N ha^{-1} (0.495).
- **Calibrated world: the dose, minus a few kilograms.** Maize at $s = 1$ loses 152.18 of 160 (0.951). Decomposition: the inaccessible half (80, lost by assumption) plus the accessible half the carbon-limited crop does not use ($80 - 7.82 = 72.18$): $80 + 72.18 = 152.18$ (my arithmetic). Part of the unused half is not lost but stays as mineral N: at the start of the 2021 season there are 68.7 kg N ha^{-1} more than at $s = 0$, i.e. 2.8% of the 16-year cumulative 2,435 kg N ha^{-1}.

- **Fixed fractions, not weather.** The 0.5 accessible fraction and the 0.5 carry-over loss (`N_CARRYOVER_LOSS`, :120) are constants. Leaching, volatilization, denitrification and N_2O are not modelled separately, so the figure ranks strategies by dose and cannot be read as an environmental loss.
- **Sign unknown.** The first draft called it a lower bound; its sign relative to reality cannot be established from this code, and the claim was removed.
- **The floor never binds.** $\max\{0, \cdot\}$ is active in 0 of 6,528 (phase, year, lattice scale) cells per world (and in none of the 1,536 raw grid nodes).

4.4 The retention trade-off

Differentiating Equations (6) and (8) in r, with $\bar{S} = \frac{1}{Tn_u}\sum_{t,u} S_{tu}$ the mean stover,

$$\frac{\partial \bar{V}}{\partial r} = -\pi_S\,\bar{S}, \qquad \frac{\partial \bar{C}}{\partial r} = +\phi\,\bar{S}, \qquad \frac{\partial \bar{V}/\partial r}{\partial \bar{C}/\partial r} = -\frac{\pi_S}{\phi} = -\frac{1000}{0.45} = -2{,}222 \text{ MXN}\,(\text{t C})^{-1}. \tag{12}$$

Both objectives are exactly linear in r and the implicit carbon price π_S/ϕ does not depend on the strategy. For the worked strategy $\bar{S} = 1.1904\,\text{t ha}^{-1}$, so going from $r = 0$ to $r = 1$ costs $1000 \times 1.1904 = 1{,}190.4\,\text{MXN ha}^{-1}$ and gains $0.45 \times 1.1904 = 0.5357\,\text{t C ha}^{-1}$:

r	$\bar{V}$ (MXN ha^{-1})	$\bar{C}$ (t C ha^{-1})	$\text{CVaR}_{0.25}$ (MXN ha^{-1})
0	11,278.0	0.1087	4,612.0
0.5	10,682.8	0.3765	4,113.1
1	10,087.6	0.6444	3,614.3

Table 10. Retention changes nothing in the simulation; it only moves stover between income and carbon.

4.5 Limiting cases and two caveats on grain and stover

- $s = 0$: $D = 0$ and $Q = Q(0)$, so $L_t = 0$ and no fertilizer is bought (test `tests/test_nsga_planner.py:85`).
- $r = 1$: no stover income; $r = 0$: carbon comes from roots alone. Both objectives are monotone in r, and $\bar{Y}, \bar{L}$ are non-decreasing in s (checked on the whole lattice in both worlds, no violation).
- $\text{CVaR}_{0.25} \le \bar{V}$ always (it is a mean of the lowest quarter); also checked on the lattice.
- Grain is dry matter but SIAP prices are on a reported (about 14% moisture) basis, so revenue is understated by about 14% for maize; second order.
- **[finding]** Default-world stover and roots are not capped like grain. The grain-N rule (`wofost_rotations.py:312--313`) caps grain, and it binds in 88% of default maize seasons at $s = 0$ and 69% at $s = 1$ (never in the calibrated world). Stover and roots are not capped, so default maize gives $15.96\,\text{t ha}^{-1}$ of stover beside $6.34\,\text{t ha}^{-1}$ of grain at $s = 1$ (all maize seasons, 2006–2021). Stover income is $14{,}639\,\text{MXN ha}^{-1}$, 92% of the default pure-maize net of 15,945 at $s = 0$. Every default-world money and carbon figure is inflated by this. Documented, not changed.

5 Dominance, the lattice, brute force and how fronts are scored

5.1 Pareto dominance

Write every strategy's objectives in minimization form $\mathbf{f}(x) = (f_1, \ldots, f_M)$, where $f_i = \sigma_i \times$ natural value (`minimized`, :226–228). Strategy x *dominates* x' when it is no worse everywhere and strictly better somewhere:

$$\mathbf{f}(x) \prec \mathbf{f}(x') \iff \forall i:\ f_i(x) \le f_i(x') \quad \wedge \quad \exists i:\ f_i(x) < f_i(x'). \tag{13}$$

The *Pareto front* is $\mathcal{P} = \{x \in \Omega : \nexists\, x' \in \Omega,\ \mathbf{f}(x') \prec \mathbf{f}(x)\}$. The code (`pareto_front`, :231–244) drops rows with identical objective vectors, then keeps the first non-dominated level of pymoo's `NonDominatedSorting`. Objectives are first rounded to 9 decimals (`FRONT_DECIMALS`), for duplicates and for dominance; compared exactly, floating-point noise of order 10^{-16} decided dominance (Section 5.4).

5.2 The lattice and brute force

A strategy lives on the lattice $\Omega = \{\omega\} \times \{0.025\,a\}_{a=0}^{50} \times \{0.05\,b\}_{b=0}^{20}$, so

$$|\Omega| = 4 \times 51 \times 21 = 4{,}284. \tag{14}$$

`N_SCALE_STEPS = 50`, `RETENTION_STEPS = 20` (:64–65); `brute_force` (:258–263) scores every point: 0.26 s in total on this machine, about 60 μs per strategy (timing varies by machine). The lattice makes the box bounds ordinary values: real-coded NSGA reaches the bounds, where these nearly linear trade-offs have their optima, only slowly. Brute force makes the exact front known, so the search can be graded.

5.3 Why the five-objective front has 483 strategies (and the default one 1,377)

Dominance resistance. If objective values were independent continuous draws, two random points are mutually non-dominated with probability

$$P(\text{mutually non-dominated}) = 1 - \frac{2}{2^M}: \quad M = 2:\ 0.50, \quad M = 3:\ 0.75, \quad M = 5:\ 0.94. \tag{15}$$

The objectives here are far from independent, but the lesson holds: with five objectives, almost nothing dominates anything. The calibrated front is 483/4,284 = 11.3% of the lattice; the default front is 1,377/4,284 = 32.1%. The 483 split into three blocks (composition computed by running `pareto_front` on the brute-force table):

Rotation	Count	s values	Why it is non-dominated
maize	$231 = 11 \times 21$	$0, 0.025, \ldots, 0.25$ (11 values, the first cell)	Each s step buys +0.00117 (t ha^{-1}) of yield at −90.7 (MXN ha^{-1}) of income and +3.27 (kg N ha^{-1}) of N loss, so no step dominates another; the 21 retentions trade carbon against income and bad-year income.
maize–barley	$231 = 11 \times 21$	same	Dominated by pure maize in the four objectives without bad-year income (all 1,071 of its strategies are). Survives only through bad-year income: −1,343 vs −4,187 (MXN ha^{-1}) for maize at $s = 0$, $r = 0$.
maize–faba bean	$21 = 1 \times 21$	0	$s = 0$ is genuinely best on income, N loss and bad-year income. (Before the float fix $s = 0.025$ also survived here through rounding noise; Section 5.4.)

Table 11. Composition of the calibrated five-objective exact front (483 strategies).

Inside the first cell $[0, 0.25]$ the calibrated yield rises only from 2.3008 to 2.3125 t ha^{-1} (+0.0117, +0.5%) for 32.7 kg N ha^{-1} more N lost and 907 MXN ha^{-1} less income. A 0.5% slope is below the model's noise, so the size of the front is a metric artifact, not decision-relevant structure.

The concrete pair: maize versus maize–faba bean.

Objective (calibrated)	maize, $s = 0.25$, $r = 0$	maize–faba bean, $s = 0$, $r = 0$
yield $\bar{Y}$ (t ha^{-1})	**2.3125**	2.0420
net income $\bar{V}$ (MXN ha^{-1})	4,988	**12,528**
N lost $\bar{L}$ (kg N ha^{-1})	32.7	**0**
carbon input $\bar{C}$ (t C ha^{-1})	**0.1104**	0.1087
bad-year income $\text{CVaR}_{0.25}$ (MXN ha^{-1})	−5,187	**5,862**

Table 12. Bold marks the winner on each objective. Each wins at least once, so by Equation (13) neither dominates the other.

Maize wins on yield (by $0.27\,\mathrm{t\,ha^{-1}}$) only as a tonnage gap: the faba bean partner yields $1.77\,\mathrm{t\,ha^{-1}}$ against maize's 2.32, yet sells for about twice the price per tonne. It also wins on carbon by a trivial $0.0017\,\mathrm{t\,C\,ha^{-1}}$. Remove yield and the maize and maize–barley blocks collapse: of the 462 non-faba members, 444 become dominated, leaving 18 (5 maize with $r \geq 0.8$ and 13 maize–barley with $r \geq 0.9$, kept by carbon input). (**Correction**: the Markdown draft said 374 dominated and 88 remaining; I could not reproduce that.) Dropping bad-year income instead gives the four-objective front of 252 (231 maize and 21 faba bean members, no maize–barley).

How big is the front at a decision-relevant resolution? (my computation; the resolutions are my illustrative choice, not repository settings.) Round each objective to a grid ε_i before testing dominance,

$$\begin{gathered} \hat{f}_i = \varepsilon_i \,\mathrm{round}\big(f_i/\varepsilon_i\big), \\ \boldsymbol{\varepsilon} = \big(0.05\ \mathrm{t\,ha^{-1}},\ 250\ \mathrm{MXN\,ha^{-1}},\ 2\ \mathrm{kg\,N\,ha^{-1}},\ 0.02\ \mathrm{t\,C\,ha^{-1}},\ 250\ \mathrm{MXN\,ha^{-1}}\big), \end{gathered} \tag{16}$$

for (yield, income, N loss, carbon, bad-year income). The calibrated five-objective front then has 22 strategies (11 maize, 9 maize–faba bean, 2 maize–barley); with the coarser $(0.1,\ 500,\ 5,\ 0.05,\ 500)$ it has 15. The 483 is therefore mostly resolution below any difference one could act on.

5.4 A floating-point artifact inside the front (found by review, fixed)

[finding] Before the fix, 21 of the 504 calibrated front members (and of the 273-member four-objective front) were maize–faba bean at $s = 0.025$, one for each retention. They were non-dominated only because their yield exceeded that at $s = 0$ by $4.4 \times 10^{-16}\,\mathrm{t\,ha^{-1}}$. The cause is Equation (2): for maize–faba bean, $X_0 = X_1$ in exact arithmetic (yield and carbon do not respond), but $(1-w)X_0 + wX_1$ is evaluated in floating point (`nsga_planner.py:142`) and lands one unit in the last place above X_0. Exact comparison in `pareto_front` then saw "strictly better in yield" and kept the point, although it is worse on income, N loss and bad-year income.

The fix. `pareto_front` (`nsga_planner.py:231--244`) now rounds the minimization-form objectives to 9 decimals before dropping duplicates and before testing dominance, and returns the rows with their unrounded values:

$$\hat{f}_i = 10^{-9}\,\mathrm{round}\big(10^{9} f_i\big), \qquad x \prec x' \text{ is tested on } \hat{\mathbf{f}}. \tag{17}$$

(`FRONT_DECIMALS = 9`, :66; tests `test_rounding_error_does_not_keep_a_strategy_on_the_front` and `test_front_rows_keep_their_unrounded_values`.) **What changed:** the calibrated five- and four-objective fronts are 483 and 252 instead of 504 and 273, the robust four-objective front is 1,727 instead of 1,730, and the final-population counts of the two calibrated NSGA-III runs fall to 118 and 77 (noise ties removed). The three-objective front (26), the default-world fronts (1,377) and the robust three-objective front (738) do not change. **One pick changed:** the calibrated sustainability-first pick on the exact front was maize–faba bean at $s = 0.025$ (maize N $= 4\,\mathrm{kg\,N\,ha^{-1}}$), one of the noise survivors; it is now $s = 0$ (N = 0), with net income 11,457 instead of 11,394 $\mathrm{MXN\,ha^{-1}}$ and N lost 0 instead of about $2\,\mathrm{kg\,N\,ha^{-1}}$. The first review had said that no pick would change; that was wrong.

5.5 IGD+ and the hypervolume ratio

To grade a found set $\mathcal{A}$ against the exact front $\mathcal{P}$, `front_quality` (:329–345) first rescales both by the reference front's range, so that $\mathcal{P}$ spans $[0,1]$ on every objective:

$$\tilde{\mathbf{f}} = \big(\mathbf{f} - \mathbf{z}^{\mathrm{id}}\big) \oslash \big(\mathbf{z}^{\mathrm{nad}} - \mathbf{z}^{\mathrm{id}}\big), \qquad z_i^{\mathrm{id}} = \min_{\mathbf{g}\in\mathcal{P}} g_i, \quad z_i^{\mathrm{nad}} = \max_{\mathbf{g}\in\mathcal{P}} g_i, \tag{18}$$

where $\oslash$ is element-wise division (the span is replaced by 1 if zero). Then, with found points $\mathbf{f} \in \mathcal{A}$, reference points $\mathbf{g} \in \mathcal{P}$ and the hypervolume reference point $\mathbf{z}^{\mathrm{ref}} = 1.1 \cdot \mathbf{1}$ (all rescaled),

$$\mathrm{IGD}^{+}(\mathcal{A}, \mathcal{P}) = \frac{1}{|\mathcal{P}|} \sum_{\mathbf{g} \in \mathcal{P}} \min_{\mathbf{f} \in \mathcal{A}} \sqrt{\sum_{i=1}^{M} \max\{0,\ f_i - g_i\}^2}, \tag{19}$$

$$\mathrm{HVR}(\mathcal{A}) = \frac{\mathrm{HV}(\mathcal{A})}{\mathrm{HV}(\mathcal{P})}, \qquad \mathrm{HV}(\mathcal{X}) = \mathrm{vol}_M\Big(\bigcup_{\mathbf{f} \in \mathcal{X}} \{\mathbf{m} :\ \mathbf{f} \le \mathbf{m} \le \mathbf{z}^{\mathrm{ref}}\} \Big). \tag{20}$$

IGD+ (inverted generational distance plus; dimensionless after rescaling) is the mean, over the reference points, of how far the nearest found point is from *weakly dominating* that reference point: a found point at least as good on every objective contributes 0. The hypervolume HV is the M-dimensional volume vol_M of the region dominated by the set and bounded by $\mathbf{z}^{\mathrm{ref}}$.

Tiny hand example ($M = 2$: N loss in $\mathrm{kg\,N\,ha^{-1}}$ and net income in $\mathrm{MXN\,ha^{-1}}$). Reference front: $(0, 1000)$, $(40, 3000)$, $(80, 5000)$. Ideal and nadir in minimization form are $(0, -5000)$ and $(80, -1000)$, so the rescaled reference points are $(0, 1)$, $(0.5, 0.5)$, $(1, 0)$. Found set $\mathcal{A}$: the two end points $(0, 1)$ and $(1, 0)$.

$$\begin{aligned} \mathrm{IGD}^{+} &= \frac{0 + \sqrt{0^2 + \max\{0, 1 - 0.5\}^2} + 0}{3} = \frac{0.5}{3} = 0.1667, \\ \mathrm{HV}(\mathcal{P}) &= \underbrace{0.5 \times 0.1}_{x \in [0, 0.5]} + \underbrace{0.5 \times 0.6}_{x \in [0.5, 1]} + \underbrace{0.1 \times 1.1}_{x \in [1, 1.1]} = 0.05 + 0.30 + 0.11 = 0.46, \\ \mathrm{HV}(\mathcal{A}) &= 1 \times 0.1 + 0.1 \times 1.1 = 0.21, \qquad \mathrm{HVR} = \frac{0.21}{0.46} = 0.4565. \end{aligned} \tag{21}$$

`front_quality` returns 0.16667 and 0.45652 on these data (run). If $\mathcal{A}$ is better than $\mathcal{P}$ the ratio exceeds 1 and IGD+ is 0 (e.g. adding $(20, 3500)$ and $(40, 4000)$ gives HVR = 1.61).

What they reward and hide. IGD+ rewards covering *every* reference point and ignores surplus points; the hypervolume ratio is dominated by the largest region, so a thin missing tail can cost little. Neither says whether a missing point matters for a decision, and both depend on the reference front, which here includes the 21 noise points above. **Naive baseline**: Table 17 compares NSGA with a uniform random sample of the same number of lattice points.

6 NSGA-II and NSGA-III in this code

Plain language. NSGA (Non-dominated Sorting Genetic Algorithm) keeps a population of candidate strategies. Each generation it breeds children by recombining and perturbing parents, then keeps the best half of parents plus children, where "best" means (a) lower in the dominance order and (b) spread out along the front. NSGA-II measures spread with a neighbor-gap statistic (crowding distance); NSGA-III, used from four objectives, measures it by how many members sit near each of a fixed set of reference directions. The code (`optimize`, :266–326) uses NSGA-II for $M \le 3$ and NSGA-III for $M \ge 4$ (:308, :315).

6.1 Encoding

A candidate is three genes: `rotation` (a categorical choice of 4), `n_step` $a \in \{0, \ldots, 50\}$ and `r_step` $b \in \{0, \ldots, 20\}$ (integers), decoded by `decode` (:292–294) to

$$s = 0.025\, a, \qquad r = 0.05\, b, \qquad f_i = \sigma_i\, f_i^{\mathrm{nat}}(\omega, s, r) \quad \text{(minimized, :305)}. \tag{22}$$

Every evaluation calls `Planner.objectives` (about 60–120 μs), so a run of 150 generations takes 3–15 s.

6.2 Non-dominated sorting and ranks

Level $\mathcal{F}_1$ is the set of non-dominated members; remove it and the non-dominated members of the rest form $\mathcal{F}_2$, and so on. Equivalently, the rank of x is the length of the longest dominance chain ending in x.

Toy point	A	B	C	D	E	F
(f_1, f_2)	$(1,5)$	$(2,3)$	$(4,1)$	$(2,5)$	$(3,4)$	$(5,5)$
Dominated by	—	—	—	A, B	B	A, B, C, D, E
Level	$\mathcal{F}_1$	$\mathcal{F}_1$	$\mathcal{F}_1$	$\mathcal{F}_2$	$\mathcal{F}_2$	$\mathcal{F}_3$

Table 13. A toy sort. pymoo's `NonDominatedSorting` returns the same levels (run).

6.3 Crowding distance

Within one level, sort the members by each objective i. A member's crowding distance is the average, over objectives, of the normalized gap between its two neighbors (boundary members get $+\infty$, so extremes are always kept):

$$\mathrm{cd}(n) = \frac{1}{M} \sum_{i=1}^{M} \frac{f_i^{(n^+)} - f_i^{(n^-)}}{f_i^{\max} - f_i^{\min}}, \tag{23}$$

where n^- and n^+ are the neighbors of n in the order sorted by f_i and the $f_i^{\max\,/\,\min}$ are taken over the level. **Toy**: points $(0,4)$, $(0.5,3)$, $(2,1)$, $(4,0)$, both ranges $= 4$.

$$\begin{aligned} \mathrm{cd}\big((0.5,3)\big) &= \tfrac{1}{2}\Big(\tfrac{2-0}{4} + \tfrac{4-1}{4}\Big) = \tfrac{1}{2}(0.5 + 0.75) = 0.625, \\ \mathrm{cd}\big((2,1)\big) &= \tfrac{1}{2}\Big(\tfrac{4-0.5}{4} + \tfrac{3-0}{4}\Big) = \tfrac{1}{2}(0.875 + 0.75) = 0.8125. \end{aligned} \tag{24}$$

The less crowded $(2,1)$ is preferred; pymoo's `calc_crowding_distance` returns the same values (run).

6.4 NSGA-II selection and survival ($M \leq 3$)

Binary tournament (`binary_tournament` with `comp_by_dom_and_crowding`, :309–314): draw two individuals; if one dominates the other it wins, otherwise the one with larger cd wins (random on ties). **Survival** (`RankAndCrowding`): merge the n_{pop} parents and n_{pop} children ($2n_{\text{pop}}$), fill the next generation level by level, and truncate the level that does not fit by decreasing cd. Population 100.

6.5 Variation on mixed variables

Code (pymoo 0.6.2, `pymoo/core/mixed.py`, read in the installed source): the two integer genes are crossed together with SBX and mutated with PM, both followed by rounding to the nearest integer; the rotation gene is crossed with uniform crossover and mutated by random choice. Duplicates (exactly equal genes) are removed.

Gene	Type	Crossover	Mutation
`n_step`, `r_step`	Integer	SBX, $\eta_c = 15$, applied with probability 0.9 per mating, then each gene crossed with probability 0.5; rounded	PM, $\eta_m = 20$, per individual 0.9, per gene 0.5; rounded
`rotation`	Choice (4)	uniform crossover: genes swapped with probability 0.5	redraw uniformly over the 4 options with probability 0.5 (mutation probability 1.0 per individual)

Table 14. Operators and their defaults as found in the pymoo source. The NSGA-III defaults (`SBX(eta=30)`) are overridden by `MixedVariableMating`.

SBX (simulated binary crossover, bounded form as implemented). For one gene with bounds $[y_{\min}, y_{\max}]$, parents $y_1 < y_2$, gap $\Delta_y = y_2 - y_1$ and a uniform draw $\zeta \in (0,1)$ shared by both children, the children y_1', y_2' are

$$\beta_L = 1 + \tfrac{2(y_1 - y_{\min})}{\Delta_y}, \quad A_L = 2 - \beta_L^{-(\eta_c+1)}, \quad \beta_q = \begin{cases} (\zeta A_L)^{\frac{1}{\eta_c+1}}, & \zeta \le 1/A_L, \\ (2 - \zeta A_L)^{-\frac{1}{\eta_c+1}}, & \text{otherwise}, \end{cases} \tag{25}$$
$$y_1' = \tfrac{1}{2}\big[(y_1 + y_2) - \beta_q\,\Delta_y\big], \qquad y_2' = \tfrac{1}{2}\big[(y_1 + y_2) + \beta_q'\,\Delta_y\big],$$

where β_q' is β_q computed from $\beta_U = 1 + 2(y_{\max} - y_2)/\Delta_y$ instead of β_L. **Toy**: parents $y_1 = 10$, $y_2 = 30$ on $[0, 50]$, $\zeta = 0.3$, $\eta_c = 15$: $\beta_L = 2$, $A_L = 1.99998$, $\beta_q = 0.96858$, so $y_1' = 10.314$ and $y_2' = 29.686$ (pymoo's own `cross_sbx` with that draw gives 10.3142 and 29.6858). After rounding the children are 10 and 30: the crossover changed nothing. By simulation of pymoo's operator, with parents 20 steps apart the children equal the parents after rounding with probability 0.55, and with parents 2 steps apart with probability 0.999. So once the population converges, SBX with $\eta_c = 15$ plus rounding is nearly a no-op and exploration comes from mutation.

PM (polynomial mutation). For a gene value y with $\xi_1 = (y - y_{\min})/(y_{\max} - y_{\min})$, $\xi_2 = (y_{\max} - y)/(y_{\max} - y_{\min})$ and a uniform draw ζ:

$$\xi_q = \begin{cases} \big[2\zeta + (1 - 2\zeta)(1 - \xi_1)^{\eta_m+1}\big]^{\frac{1}{\eta_m+1}} - 1, & \zeta \le \tfrac{1}{2}, \\ 1 - \big[2(1-\zeta) + 2(\zeta - \tfrac{1}{2})(1 - \xi_2)^{\eta_m+1}\big]^{\frac{1}{\eta_m+1}}, & \zeta > \tfrac{1}{2}, \end{cases} \qquad y' = y + \xi_q\,(y_{\max} - y_{\min}). \tag{26}$$

Toy: $y = 10$ on $[0, 50]$, $\eta_m = 20$, $\zeta = 0.2$: $\xi_1 = 0.2$, $(0.8)^{21} = 0.00922$, bracket $= 0.4 + 0.6 \times 0.00922 = 0.40553$, $\xi_q = 0.40553^{1/21} - 1 = -0.04207$, $y' = 10 - 0.04207 \times 50 = 7.897 \to 8$ (pymoo: 7.8966). Over many draws from $y = 10$ the rounded move has mean size 2.2 steps; 19% of mutations round back to 10 and 66% stay within ± 2.

6.6 NSGA-III: reference directions, association, niching

Das–Dennis reference directions. With M objectives and p partitions per axis, the reference directions are the points of the unit simplex whose coordinates are multiples of $1/p$:

$$\mathcal{W} = \Big\{ \boldsymbol{\psi} = \tfrac{1}{p}(k_1, \ldots, k_M) : \; k_i \in \mathbb{N}_0, \; \sum_{i=1}^{M} k_i = p \Big\}, \qquad H = |\mathcal{W}| = \binom{p + M - 1}{M - 1}. \tag{27}$$

For example $M = 3$, $p = 3$ gives $\binom{5}{2} = 10$ directions, from $(0, 0, 1)$ through $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ to $(1, 0, 0)$. The three cases used here (`optimize`, :316–317) are, with the population size defaulting to H:

Run	M	p	$H = \binom{p+M-1}{M-1}$	Population	Where
4 objectives (default)	4	6	$\binom{9}{3} = 84$	84	`{4: 6, 5: 5}` at :316
5 objectives (default)	5	5	$\binom{9}{4} = 126$	126	same
robust, 4 objectives	4	10	$\binom{13}{3} = 286$	286	`--partitions 10`

Table 15. Reference directions (counts confirmed by calling pymoo's `get_reference_directions`).

Normalization, association, niching (`ReferenceDirectionSurvival`, read in the installed pymoo). Each generation the objectives are mapped to $\tilde{f}_i = (f_i - z_i^{\mathrm{id}})/(a_i - z_i^{\mathrm{id}})$, with z^{id} the running ideal point and a_i the intercepts of the hyperplane through the extreme points (falling back to the worst value on the first level). Each member is associated with the reference ray at the smallest perpendicular distance

$$d_\perp(\tilde{\mathbf{f}}, \boldsymbol{\psi}) = \Big\| \tilde{\mathbf{f}} - \frac{\boldsymbol{\psi}^\top \tilde{\mathbf{f}}}{\boldsymbol{\psi}^\top \boldsymbol{\psi}}\, \boldsymbol{\psi} \Big\|_2, \qquad h(\tilde{\mathbf{f}}) = \arg\min_{\boldsymbol{\psi} \in \mathcal{W}} d_\perp(\tilde{\mathbf{f}}, \boldsymbol{\psi}), \tag{28}$$

and each ray ψ_h has a niche count ν_h (members of the already accepted levels associated with it). The splitting level is filled by repeatedly picking the niche with the smallest ν_h among those that still have candidates: if $\nu_h = 0$ take the candidate with the smallest $d_\perp$, otherwise a random one. **Toy** ($M = 2$, $p = 2$, so $H = 3$ rays $(1,0)$, $(\frac{1}{2},\frac{1}{2})$, $(0,1)$; point $\tilde{\mathbf{f}} = (0.6, 0.2)$, $|\tilde{\mathbf{f}}|^2 = 0.40$):

$$\begin{aligned} d_\perp(\cdot,(1,0)) &= \sqrt{0.40 - 0.6^2} = 0.200, \qquad d_\perp(\cdot,(\tfrac{1}{2},\tfrac{1}{2})) = \sqrt{0.40 - \left(\tfrac{0.8}{\sqrt{2}}\right)^2} = \sqrt{0.08} = 0.283, \\ d_\perp(\cdot,(0,1)) &= \sqrt{0.40 - 0.2^2} = 0.600, \end{aligned} \tag{29}$$

so the point belongs to the first niche. Selection for mating is a random tournament (`comp_by_cv_then_random`: with no constraints it picks one of the two at random), so all selection pressure comes from survival.

Why NSGA-III from four objectives. With many objectives most of a population is mutually non-dominated, so ranks stop discriminating and a neighbor-gap statistic stops being meaningful. Evidence in this repository: in all five NSGA-III runs of Table 16 the final population is almost entirely non-dominated (84, 126 and 286 members for the default and robust runs; 77 and 118 of 84 and 126 for the calibrated runs once noise ties are removed), and on the test fixture all 462 strategies are mutually non-dominated (`tests/test_nsga_planner.py:199--204`). That the replacement works better than NSGA-II would here is the standard argument (Deb and Jain, 2014); I did not run NSGA-II with $M \geq 4$. The normalization looked sound on the calibrated four-objective run (100 generations, seed 1): the ideal point matched the exact front's exactly and the nadir was close (36.6 against 32.7 kg N ha^{-1} for N loss).

6.7 The reporting bug that was fixed: `result.X` versus the final population

Code, before. `optimize` returned pymoo's `result.X`. For NSGA-III, pymoo's `_set_optimum` sets this to `survival.opt`: the first-level members that are, for some *occupied* reference direction, the closest to that ray. Hence

$$\left|\texttt{result.X}\right| \;\leq\; \left|\left\{h(\tilde{\mathbf{f}}_n) :\; n \in \text{merged population}\right\}\right| \tag{30}$$

(the number of occupied niches, i.e. reference rays that are the nearest ray of at least one member), regardless of how many non-dominated strategies the population holds. *Code, now* (:323–326): `result.pop` is decoded and reduced to its non-dominated set. Re-running the documented commands (seed 1):

		`result.X` (before)		final population (now)		all evaluated	
Run (generations, H)	Exact	count	HVR	count	HVR	unique	HVR
calibrated 4 obj (200, 84)	252	18	0.937	77	0.981	1,485	1.000
calibrated 5 obj (150, 126)	483	16	0.888	118	0.985	1,624	1.000
default 4 obj (200, 84)	1,377	23	0.908	84	0.954	3,341	1.000
default 5 obj (150, 126)	1,377	22	0.925	126	0.965	3,248	0.999
robust 4 obj (150, 286)	1,727	45	0.927	286	0.977	4,246	1.000

Table 16. Coverage by hypervolume ratio (Equation (20)) before and after the fix. "All evaluated": non-dominated set of every distinct lattice point the run evaluated. The "now" columns equal the saved `run.json` values (e.g. 0.9850, 0.9809, 0.9544).

The count returned equals the number of occupied niches in every run (18/84, 16/126, 23/84, 22/126, 45/286; I read the niche of each final member), so 4–8 population members pile into each occupied niche. The search was never deficient: the final population already holds 95–98.5% of the hypervolume, and the archive of everything evaluated recovers essentially the whole front. What limits the returned front is the population cap (77–286 returned against fronts of 252–1,727; the two calibrated counts are below the population size because noise ties are removed). IGD+ before → now: 0.016 → 0.006, 0.018 → 0.005, 0.048 → 0.029, 0.061 → 0.030, 0.037 → 0.016.

NSGA-II on three objectives. It is exact when the front fits in the population (26 of 26, seeds 1–3, IGD+ 0) and otherwise keeps 0.965–0.973 of the hypervolume (default, 1,377 exact points, population 100; IGD+ 0.023–0.025).

6.8 Is the search worth it at this size? A naive baseline

Run	Evaluated	Returned	Archive	Random	Exact
calibrated 4 obj	1,485 (35% of 4,284)	0.981	1.000	0.9665	252
calibrated 5 obj	1,624 (38%)	0.985	1.000	0.9661	483
default 4 obj	3,341 (78%)	0.954	1.000	0.9983	1,377
default 5 obj	3,248 (76%)	0.965	0.999	0.9985	1,377
robust 4 obj	4,246 (99%)	0.977	1.000	0.9999	1,727

Table 17. Hypervolume ratio for NSGA's *returned* final population, NSGA's *archive* (everything evaluated), and a uniform random sample of the same number of lattice points ("Random": mean of 10 draws, my run); "Exact" is the exact front size. The returned set is capped by the population size, a random sample's is not, so compare the archive with the random column.

NSGA beats random sampling at equal evaluation count only where the front is small (calibrated), and it visits 35–99% of the lattice, so against brute force (all 4,284 in 0.26 s) it saves nothing. Its role here is a check of the machinery, which would earn its keep only in a much larger decision space. **Why a lattice** (versus the earlier real-coded genes). I re-ran an emulation of the real-coded variant: on the test fixture (population and generations 40) IGD+ was 0.098, 0.172, 0.133 for seeds 1–3; on the real data (population 100, 100 generations) 0.010–0.011 (calibrated) and 0.018–0.020 (default), against 0.000 and 0.023–0.025 on the lattice. The lattice matters at small budgets and is a wash on the real problem.

7 The compromise picker `choose`: weighted Chebyshev

Plain language. A front of hundreds of strategies is not a decision. `choose` (:348–358) returns the front member that is closest to the "utopia" point (best value of every objective on that front) after each objective is rescaled to $[0, 1]$ and weighted. Distance is the *largest* weighted shortfall, so the pick is the strategy whose worst weighted objective is least bad.

7.1 The equation

For a front with members $n = 1, \ldots, |\mathcal{P}|$ and minimization-form values f_{ni},

$$\tilde{f}_{ni} = \frac{f_{ni} - \min_{n'} f_{n'i}}{\max_{n'} f_{n'i} - \min_{n'} f_{n'i}} \in [0,1], \qquad \lambda_i = \frac{v_i}{\sum_{i'} v_{i'}},$$
$$\gamma_n = \max_i \ \lambda_i \tilde{f}_{ni}, \qquad n^* = \arg\min_n \gamma_n. \tag{31}$$

Here $\tilde{f}_{ni}$ is a dimensionless shortfall (0 = best on the front, 1 = worst), v_i are the user's weights (:85–88), λ_i their normalization and γ_n is the score (smaller is better). Code: the whole of Equation (31) is the single expression at :358, with the span replaced by 1 when an objective is constant on the front (:356). The ranges come from *the front that is passed in.*

Toy. Front (yield, N loss) = $(3, 100)$, $(2, 50)$, $(1, 0)$, weights 0.6 (yield) and 0.4 (N loss); yield is maximized, so $f_1 = -3, -2, -1$:

$$\tilde{\mathbf{f}} = \begin{pmatrix} 0 & 1 \\ 0.5 & 0.5 \\ 1 & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} \max(0.6 \cdot 0,\ 0.4 \cdot 1) \\ \max(0.6 \cdot 0.5,\ 0.4 \cdot 0.5) \\ \max(0.6 \cdot 1,\ 0.4 \cdot 0) \end{pmatrix} = \begin{pmatrix} 0.4 \\ 0.3 \\ 0.6 \end{pmatrix} \ \Rightarrow\ n^* = 2 \text{ (the middle one).} \tag{32}$$

`choose` returns index 1 (zero-based), as it should (also `tests/test_nsga_planner.py:290--295`).

Assumptions. A Chebyshev score can reach non-convex parts of a front, unlike a weighted sum, and the weights are importance judgments (0.30 carbon, 0.20 bad-year income, 0.20 N loss, 0.20

income, 0.10 yield for "sustainability first"). But it uses only the single binding objective of each member.

7.2 Why picks are unstable

1. **Only the binding objective counts.** On the exact robust four-objective front (1,727 strategies), "sustainability-first" scores 0.1065 at maize N = 128, 0.1066 at 124, 0.1067 at 120 and 0.1082 at 88 kg N ha^{-1}, all bound by the carbon shortfall; 36 members score within 0.01 of the best, spanning 0–140 kg N ha^{-1}. The dose is decided in the fourth decimal.
2. **Ties are broken by row order.** In "profit-first" on the same front, 547 members tie at exactly the minimum score 0.2462 because the same yield term binds for all of them, and `np.argmin` returns the first row (maize N = 4 kg N ha^{-1}).
3. **Ranges come from the front itself.** The pick moves when the front is sampled instead of exact (Table 19).
4. **Range normalization amplifies trivia.** On the calibrated exact front the yield range is 2.042–2.3125 t ha^{-1} and income 1,824–12,528 MXN ha^{-1}:

Strategy		income	yield	bad-year	N loss	carbon
	weight λ_i	0.40	0.30	0.10	0.10	0.10
maize, $s = 0$, $r = 0$	$\tilde{f}_{ni}$	0.620	0.043	0.813	0	0.998
(income 5,895)	$\lambda_i \tilde{f}_{ni}$	0.248	0.013	0.081	0	0.100
	$\gamma_n = \max$	**0.248** (income binds)				
maize–faba bean, $s = 0$, $r = 0$	$\tilde{f}_{ni}$	0	1.000	0	0	1.000
(income 12,528)	$\lambda_i \tilde{f}_{ni}$	0	0.300	0	0	0.100
	$\gamma_n = \max$	0.300 (yield binds)				

Table 18. Why "profit-first" picks pure maize on the calibrated exact front. For maize–faba bean, $\tilde{f}_{n,\text{yield}} = (2.3125 - 2.042)/(2.3125 - 2.042) = 1$.

"Profit-first" gives up $1 - 5{,}895/12{,}528 = 53\%$ of attainable income to avoid a 0.26 t ha^{-1} tonnage gap that is mostly an artifact of averaging maize and faba bean tonnes.

Front	Weights	Exact-front pick	NSGA-front pick
calibrated (5 obj)	sustainability	mf, $N = 0$, $r = 0.90$	mf, $N = 0$, $r = 0.95$
	profit	mz, $N = 0$, $r = 0$	mz, $N = 0$, $r = 0$
default (5 obj)	sustainability	mf, $N = 136$, $r = 1.00$	mf, $N = 124$, $r = 1.00$
	profit	mf, $N = 156$, $r = 0.75$	mf, $N = 152$, $r = 0.75$
robust (4 obj)	sustainability	mf, $N = 128$, $r = 1.00$	mf, $N = 104$, $r = 1.00$
	profit	mf, $N = 4$, $r = 0$ (first of 547 ties)	mf, $N = 8$, $r = 0.10$

Table 19. Compromise picks; mf = maize–faba bean, mz = maize, N = maize N in kg N ha^{-1}. Exact fronts: the saved `brute_force_front.csv`; NSGA: the saved `front.csv`.

Takeaway. Use the picks to see the trade-offs, not as findings; in a space this small choose from the brute-force front, and report the shortlist around the pick.

8 The robust planner `RobustPlanner`

Plain language. The default and calibrated parameter sets disagree completely about what nitrogen buys, and the default world's money is about 8× larger. The robust planner scores a strategy in *both* worlds, measures in each world how far it is from the best attainable value of that objective as a share of that world's range, and keeps the *worse* of the two. A strategy is robust if it is never far from the best in either world (a minimax-regret flavour).

8.1 The equations

Let $f_i^{(\theta)}(x)$ be objective i of strategy x in world $\theta \in \{\text{def}, \text{cal}\}$, in minimization form. Over the lattice Ω of that world,

$$\begin{aligned} z_i^{(\theta)} &= \min_{x' \in \Omega} f_i^{(\theta)}(x'), \qquad \Delta_i^{(\theta)} = \max_{x' \in \Omega} f_i^{(\theta)}(x') - z_i^{(\theta)}, \\ \delta_i^{(\theta)}(x) &= \frac{f_i^{(\theta)}(x) - z_i^{(\theta)}}{\Delta_i^{(\theta)}}, \quad \rho_i(x) = \max_{\theta} \ \delta_i^{(\theta)}(x). \end{aligned} \tag{33}$$

$\delta_i^{(\theta)} \in [0, 1]$ is the shortfall (dimensionless; 0 = the best attainable in that world, 1 = the worst) and ρ_i is the robust objective, always minimized (named `shortfall_<objective>`). Code: z is `self.best[world]` (:211) and Δ is `self.span[world]` (:212, set to 1 if constant); the division is at :220–221 and the maximum over worlds at :222; the lattice values come from `brute_force` per world (:210). The same search machinery then runs on $\rho_1, \ldots, \rho_M$. Tests: `tests/test_nsga_planner.py:224--246`.

8.2 A worked strategy in both worlds

Strategy: maize–faba bean, $s = 0.55$ (maize N = $88\,\mathrm{kg\,N\,ha^{-1}}$), $r = 0$. World constants: net income best/span (def) 106,995.47 / 105,689.30 and (cal) 12,528.02 / 13,454.44 $\mathrm{MXN\,ha^{-1}}$.

$$\begin{aligned} \delta^{\text{def}}_{\text{income}} &= \frac{106{,}995.47 - 95{,}902.38}{105{,}689.30} = \frac{11{,}093.09}{105{,}689.30} = 0.1050, \\ \delta^{\text{cal}}_{\text{income}} &= \frac{12{,}528.02 - 11{,}153.02}{13{,}454.44} = \frac{1{,}375.00}{13{,}454.44} = 0.1022, \\ \delta^{\text{def}}_{\text{N loss}} &= \frac{27.276 - 0}{98.940} = 0.2757, \qquad \delta^{\text{cal}}_{\text{N loss}} = \frac{54.720 - 0}{192.178} = 0.2847. \end{aligned} \tag{34}$$

Objective	default value	δ^{def}	calibrated value	δ^{cal}	$\rho = \max$ (world)
net income ($\mathrm{MXN\,ha^{-1}}$)	95,902	0.1050	11,153	0.1022	0.1050 (def)
N loss ($\mathrm{kg\,N\,ha^{-1}}$)	27.3	0.2757	54.7	0.2847	0.2847 (cal)
yield ($\mathrm{t\,ha^{-1}}$)	7.12	0.2330	2.04	0.7808	0.7808 (cal)
carbon input ($\mathrm{t\,C\,ha^{-1}}$)	1.07	0.9859	0.109	1.0000	1.0000 (cal)
bad-year income ($\mathrm{MXN\,ha^{-1}}$)	87,750	0.0090	4,487	0.0840	0.0840 (cal)

Table 20. The five shortfalls of the worked strategy (code output matches the hand values in the first two rows exactly).

8.3 Why range normalization, and its weaknesses

The worlds' money differs about 8× (best income 106,995 versus 12,528 $\mathrm{MXN\,ha^{-1}}$), so raw regret in pesos would let the default world decide everything; dividing by each world's range puts both on $[0, 1]$. But:

- **Narrow ranges inflate trivia.** The calibrated yield span is $0.346\,\mathrm{t\,ha^{-1}}$ (1.966–2.312) against 6.78 in the default world, so any change of rotation reads as a huge shortfall (0.78 wherever faba bean is in the rotation). Leave yield out of the robust objectives unless tonnage matters.
- **Spans include dominated corners.** The calibrated N-loss span is $192\,\mathrm{kg\,N\,ha^{-1}}$ (maize at $s = 1.25$, $200\,\mathrm{kg\,N\,ha^{-1}}$ dose), while the worst N loss on the exact front is 32.7. N-loss shortfalls therefore depend on the arbitrary grid upper bound of 1.25.
- **The ideal point is per objective**, so it is not jointly attainable; and weighing the two worlds equally is a modelling choice, not a probability.

8.4 The minimax dose, derived

Fix the rotation (maize–faba bean) and $r = 0$, and vary maize N $= N$. Let $\bar{V}_\theta^*$ be the best mean income in world θ and $R_\theta(N) = \bar{V}_\theta^* - \bar{V}_\theta(N)$ the regret in pesos (MXN ha^{-1}). Whatever the normalizer κ_θ (range, best income, or 1), the income shortfall is R_θ/κ_θ.

- **Calibrated world:** yield does not respond, so income is exactly linear in the dose. The phase-average dose is μN with $\mu = (1 + \frac{40}{160})/2 = 0.625$, hence $R_{\text{cal}}(N) = m_{\text{cal}} N$ with $m_{\text{cal}} = \pi_N \mu = 25 \times 0.625 = 15.625$ MXN (kg N)$^{-1}$ (check: at 88, $15.625 \times 88 = 1{,}375.0$, as above).
- **Default world:** income rises with the dose and is best at $N = 200$, so the regret falls from $R_{\text{def}}(0)$ to 0 at an average slope m_{def} (cell slopes 85 to 107):

$$R_{\text{def}}(0) = 106{,}995.47 - 87{,}966.19 = 19{,}029.28 \text{ MXN ha}^{-1},$$
$$m_{\text{def}} = \frac{19{,}029.28}{200} = 95.1 \text{ MXN (kg N)}^{-1}. \tag{35}$$

The robust income shortfall is the larger of a falling and a rising line, so it is smallest where they cross:

$$\frac{R_{\text{def}}(0) - m_{\text{def}} N^*}{\kappa_{\text{def}}} = \frac{m_{\text{cal}} N^*}{\kappa_{\text{cal}}} \quad \Longrightarrow \quad N^* = \frac{R_{\text{def}}(0)}{m_{\text{def}} + m_{\text{cal}}\,\Lambda}, \qquad \Lambda = \frac{\kappa_{\text{def}}}{\kappa_{\text{cal}}}. \tag{36}$$

Only the ratio Λ matters: it is how many calibrated-world pesos one default-world peso is worth. With the repository's range normalization, $\Lambda = 105{,}689.30/13{,}454.44 = 7.855$:

$$N^* = \frac{19{,}029.28}{95.15 + 15.625 \times 7.855} = \frac{19{,}029.28}{95.15 + 122.73} = \frac{19{,}029.28}{217.88} = 87.3 \text{ kg N ha}^{-1}. \tag{37}$$

Solving the same equation on the exact piecewise-linear curves gives 89.4; on the lattice (4 kg steps) the worst shortfall is 0.1084 at 84, 0.1050 at 88 and 0.1068 at 92, so the minimax dose is 88. *Beyond* N^*, the calibrated income shortfall rises, and N loss rises with dose in both worlds, so those doses are dominated on (income, N loss): the two-objective robust front has 23 strategies, all maize–faba bean at $r = 0$ with maize N from 0 to 88 kg N ha^{-1} $(0, 4, \ldots, 88)$.

8.5 The same dose for every rotation, and its dependence on how regret is defined

Rotation ($r = 0$)	N^*	cal. $\bar{V}$	cal. $\bar{L}$	def. $\bar{V}$	def. $\bar{L}$
maize	100	3,488	92.2	36,375	49.5
maize–barley	84	2,375	54.4	26,989	28.6
maize–faba bean	88	11,153	54.7	95,902	27.3
maize–barley–faba bean	72	7,905	38.8	70,339	19.4

Table 21. Minimax maize N N^* (kg N ha^{-1}) on the lattice (range-normalized income shortfall) and the income $\bar{V}$ (MXN ha^{-1}) and N lost $\bar{L}$ (kg N ha^{-1}) there, in each world. Reproduced from the code; matches `NSGA_RESULTS.md` on every row. For maize–faba bean at 88, income is 89% (cal) and 90% (def) of the best attainable.

Normalizer κ_θ	Λ	closed form	exact crossing	lattice, mf	lattice, mz / mb / mbf
range of income (the repo)	7.855	87.3	89.4	88	100 / 84 / 72
share of the best income	8.540	83.2	85.1	84	84 / 60 / 56
absolute pesos ($\kappa = 1$)	1	171.8	172.7	176	200 / 200 / 200

Table 22. Minimax maize N (kg N ha^{-1}) by normalizer, through Λ in Equation (36); mf = maize–faba bean, mz = maize, mb = maize–barley, mbf = maize–barley–faba bean. In absolute pesos the other three rotations sit at the grid's upper bound (200 kg N ha^{-1}).

The hedge is a property of the normalizer. Equation (36) makes it explicit: a larger Λ (the calibrated world counts for more) lowers the dose, from 172 at $\Lambda = 1$ to 83 at $\Lambda = 8.5$. Absolute

pesos make the default world's larger money decide, so the "hedge" simply follows the default model. Prices matter as well: over N price $\in \{10, 25, 50\}$, faba bean price $\times\{0.5, 0.75, 1\}$ and stover price $\in \{0, 500, 1000\}$ MXN t^{-1}, the lattice minimax dose for maize–faba bean ranges from 48 to 132 kg N ha^{-1} (maize alone: 100 to 148). The extremes are 48 (N price 50, faba bean at half price, stover 1,000) and 132 (N price 10, faba bean at full price); 27 re-runs of the planner, my script.

The statement that survives is therefore: "below the current 160 and above 0 under range normalization and these placeholder prices". It is a hedge by construction, not a finding about nitrogen.

9 Statistics and honesty check

9.1 The holdout (2022–2025)

The planner is built twice from the same table: a design `Planner` on 2006–2021 and a holdout `Planner` on 2022–2025 (`load_planners`, :482–488). The saved fronts carry the holdout values as `*_holdout` columns. For the two compromises picked from the calibrated NSGA front:

		design 2006–2021		holdout 2022–2025	
Pick (calibrated NSGA front)	(s, r)	$\bar{V}$	$\text{CVaR}_{0.25}$	$\bar{V}$	CVaR ($q = 1$)
sustainability-first (maize–faba bean)	(0, 0.95)	11,397	4,914	11,598	8,843
profit-first (maize)	(0, 0)	5,895	−4,187	6,158	2,814

Table 23. Net income and bad-year income in MXN ha^{-1}. With four holdout years $q = \lceil 0.25 \times 4 \rceil = 1$, so the holdout "bad-year" value is the single worst year.

The order of the two picks is kept. But the holdout does not test resilience: 2023 had 339.5 mm of season rain, below every design year (range 558–887, mean 727), yet the calibrated model still gives 1.99 t ha^{-1} of maize (design mean 2.30), so the calibrated model barely responds to drought. (The Markdown draft quoted 11,598 against 6,086: the sustainability-first pick is back at 11,598 after the float fix, and the profit-first pick of the current front holds out at 6,158.) The parameters were fit to 2006–2021 SIAP yields; 2022–2025 never selected the knobs but was looked at while the first calibration was designed (`CALIBRATION_RESULTS.md`).

9.2 The CVaR over 16 years

- **It is the mean of four numbers** (Equation (11)): descriptive, with no confidence interval in the code. Illustration (my computation): an i.i.d.-year bootstrap of the worked strategy's 16 incomes (20,000 resamples, seed 0) gives a 5th–95th percentile range [439, 8,220] MXN ha^{-1} for $\text{CVaR}_{0.25}$ (point value 4,113) against [8,598, 12,544] for the mean $\bar{V}$: the tail statistic is several times noisier.
- **Bad years are frost years, not drought years, in the calibrated world.** Among the 16 pure-maize seasons at $s = 0$, the point-biserial correlation of yield with a frost-ended season is −0.70 and with season rain −0.02 (aggregated by year over all crops: −0.65 and −0.19). With $n = 16$ the Fisher-z 95% interval for −0.70 is roughly $[-0.89, -0.31]$ (my arithmetic). *Docs* (`CALIBRATION_RESULTS.md`): the POWER grid cell sits ≈ 380 m above the fields and is about 2.5°C colder, which I did not re-derive. So "bad-year income" measures frost truncation and sowing date.

9.3 Real data, placeholders, simulated, proxied

Category	Items
Real data	POWER weather; IGBP-DIS and GLDAS soils; SIAP prices (median of three years) and the observed yields behind the calibration. But the faba bean price rests on 10–12 ha of faba bean per year, and its latest value (5,030 in 2025) is 0.30× the median used.
Placeholders	Costs 14,000 / 9,000 / 11,000 MXN ha^{-1}, N price 25, stover price 1,000, carbon fraction 0.45 (code comment: typical 0.40–0.47).
Simulated (WOFOST)	Grain yield, N uptake, stover and root biomass, year by year, for chained seasons.
Proxied or accounting	*Soil carbon*: an input flow, no decomposition, no stock. *N loss*: accounting against hand-set fractions. *Retention*: no feedback on yield, N or water. *Prices*: constant. No N_2O or fertilizer-manufacturing emissions.

9.4 Sensitivity of the exact-front picks

Re-running `choose` on the exact fronts after one change at a time ("mf" is maize–faba bean, "mz" maize, N the maize dose in kg N ha^{-1}; each cell reads sustainability-first / profit-first):

Change	Calibrated (front size)	Default (front size)
as shipped	mf, $N = 0$, $r = 0.90$ / mz, $N = 0$, $r = 0$ (483)	mf, $N = 136$, $r = 1.00$ / mf, $N = 156$, $r = 0.75$ (1,377)
N price 0	unchanged / mz, $N = 40$, $r = 0$ (483)	unchanged
N price 100	unchanged	unchanged / mf, $N = 148$, $r = 0.75$
faba price 5,030 (SIAP 2025)	mz, $N = 0$, $r = 0.55$ / mz, $N = 0$, $r = 0.35$ (462)	unchanged / mf, $N = 200$, $r = 0.35$ (1,517)
stover price 0	both mz, $N = 0$, $r = 1$ (23)	mz, $N = 160$, $r = 1$ / mf, $N = 200$, $r = 1$ (102)
carbon fraction 0.40	unchanged	unchanged (it cancels after range normalization)

Table 24. One-at-a-time changes. The nitrogen response itself (calibrated versus default) changes every pick more than any price does; only a farm fertilizer trial can settle it.

10 Verification log, findings and what to keep in mind

10.1 What I ran and what held

- `python -m unittest tests.test_nsga_planner`: 35 tests OK (5.4 s).
- Independent pandas recomputation from the raw `grid.csv` (10 strategies, both worlds, interpolated and node s): maximum relative difference 1.2×10^{-16}.
- Worked strategy, node tables and leave-one-node-out error (Section 4, Tables 9, 5, 6): hand and code agree.
- Front sizes 483, 1,377, 252, 26, 738, 1,727 recomputed (after the float fix); the saved `brute_force_front.csv` sets equal the recomputed sets (calibrated, default).
- NSGA re-runs, seed 1, documented commands: hypervolume ratios equal the saved `run.json` (0.9850, 0.9809, 0.9544, …); grid hash `df8bfe11...` matches.
- Minimax table, price sweep and three normalizers; picks on exact and NSGA fronts; sensitivity table; four-plan cross-table: reproduced.
- Toys (sorting, crowding, SBX, PM, IGD+, HVR, perpendicular distance): checked against pymoo and `front_quality`.
- Monotonicity over the whole lattice, both worlds: $\bar{Y}, \bar{L}$ non-decreasing in s; $\bar{V}$ non-increasing and $\bar{C}$ non-decreasing in r; CVaR $\leq \bar{V}$; no violation.
- Random-sample baseline, bootstrap and correlations: run (Table 17, Section 9).

10.2 What I could not verify

- `--build-grid`: needs PCSE and a NASA login; the saved grid was taken as given (its hash matches the runs; `grid.json` records commit `21e30ec`).

- The SMAP correlation $r = 0.87$ (needs a simulation) and which parameter set produced it.
- The earlier real-coded implementation (never committed; I emulated it), and standard NSGA-II/III theory (quoted from memory; those papers are not in `Articles/`).
- That the "N lost" figure is physically meaningful, and that default faba bean at $9.8\,\mathrm{t\,ha^{-1}}$ is plausible (it is not, against 1–2 $\mathrm{t\,ha^{-1}}$ typical in Mexico; *docs*).

10.3 Findings

1. **[finding]** Floating-point survivors in the calibrated fronts: 21 strategies existed only through a $4.4 \times 10^{-16}\,\mathrm{t\,ha^{-1}}$ yield difference and one pick depended on them. *Fixed* (Section 5.4).
2. **[finding]** Ties in `choose` are resolved by row order (:358, `np.argmin`); 547 robust-front members tie exactly in profit-first (Section 7).
3. **[finding]** Default-world stover and roots are uncapped (`wofost_rotations.py:312--318`): grain is capped by the grain-N rule but stover is not (Section 4). Open.
4. *Minor documentation slips*, fixed in the repository documents: the results document said "4–12 ha of faba bean" (it is 10–12 ha; 4–12 is barley and faba bean together), and a code comment said "16 of 84" (now "16–18 of 84").

10.4 Things to keep in mind

1. The calibrated "zero N" and the default "85–100% of the dose" are properties of two parameter sets, not agronomy; the fertilizer decision is the one input the data cannot settle.
2. Do not read the 483- and 1,377-point fronts or the Chebyshev picks as findings: they come from tiny slopes (0.5% yield), dominance resistance and range normalization. At a decision-relevant resolution the front has about 15–22 strategies.
3. The "50–130 $\mathrm{kg\,N\,ha^{-1}}$ hedge" exists only under range-style regret and the placeholder prices (Table 22).
4. Everything monetary in the default world is driven by uncapped stover and an implausible faba bean yield; the bad-year objective measures frost and sowing date, not drought.

A check you could run next. Redo the resolution-rounded front of Section 5 with the smallest income and yield differences a farmer would act on, and see which rotations and doses survive; then ask what interior grid nodes at $s = 0.05, 0.1, 0.15$ (`--build-grid` with `GRID_SCALES` edited; needs PCSE) say about the response inside the first cell, where the calibrated front's maize strategies sit.

Background references (quoted from memory; not in the repository): Deb et al. (2002), *IEEE TEVC* 6(2), NSGA-II; Deb and Jain (2014), *IEEE TEVC* 18(4), NSGA-III; Das and Dennis (1998), *SIAM J. Optim.* 8(3); Ishibuchi et al. (2015), IGD+; Rockafellar and Uryasev (2000), *J. Risk* 2(3), CVaR.